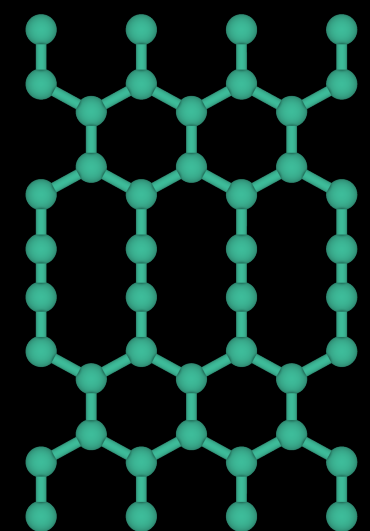


# On the CO<sub>2</sub> Harvesting from N<sub>2</sub> Using Grazyne Membranes

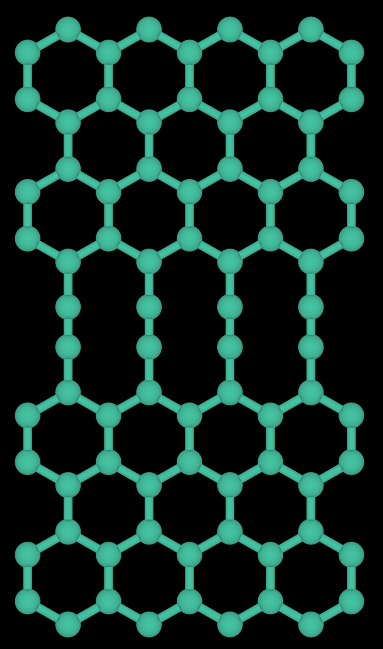
## WE PROPOSE GRAZYNES

Grazynes are a subtype of graphynes, 2D single-layer C-based materials with sp and sp<sup>2</sup> C atoms hybridizations. In grazynes, graphene-like stripes with sp<sup>2</sup> hybridized C atoms are interconnected by acetylenic linkages with sp C atoms.

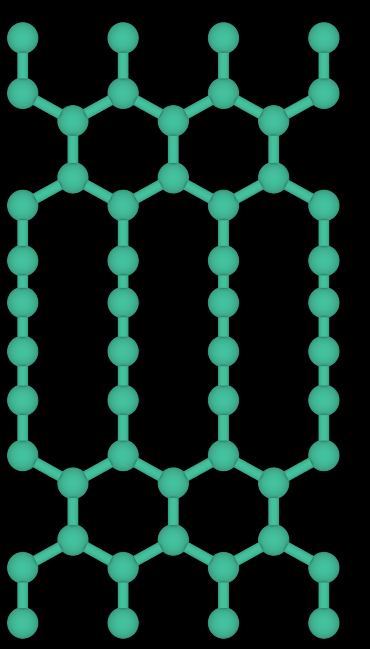


[1],[1]-grazyne

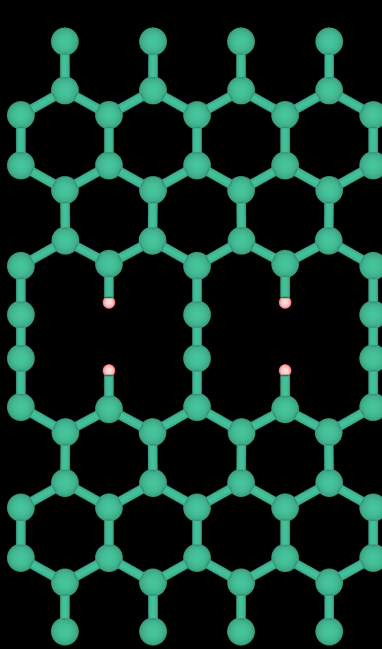
They can be modified in different ways



Modifying the graphene width

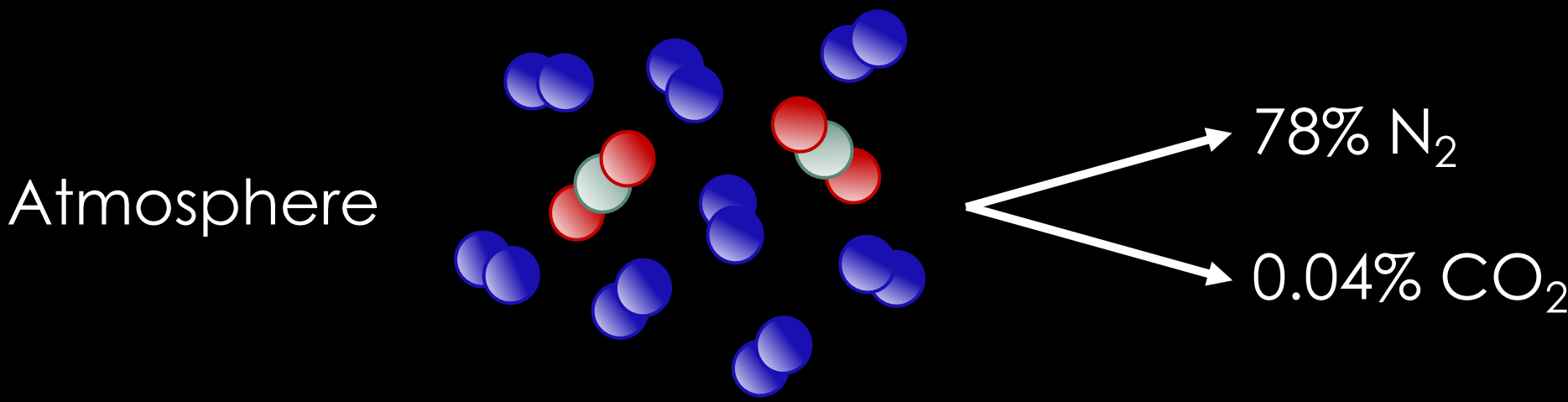


Enlarging acetylenic linkages



Creating acetylenic vacancies

The increase in the concentration of CO<sub>2</sub> in the atmosphere is one of the main drivers of climate change. In this context, the separation of CO<sub>2</sub> from N<sub>2</sub> is a critical process.

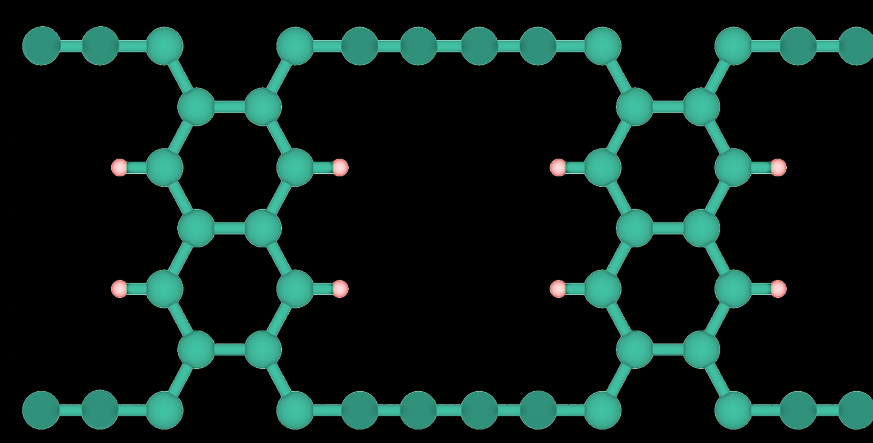


To effectively use this N<sub>2</sub>, it must be in a high-purity state, which requires purification processes. Within the existing options, membrane-based separation stands out as a promising solution for CO<sub>2</sub> separation, boosting numerous benefits such as environmental friendliness, a high active surface area, and easy maintenance.

Air Cleaner

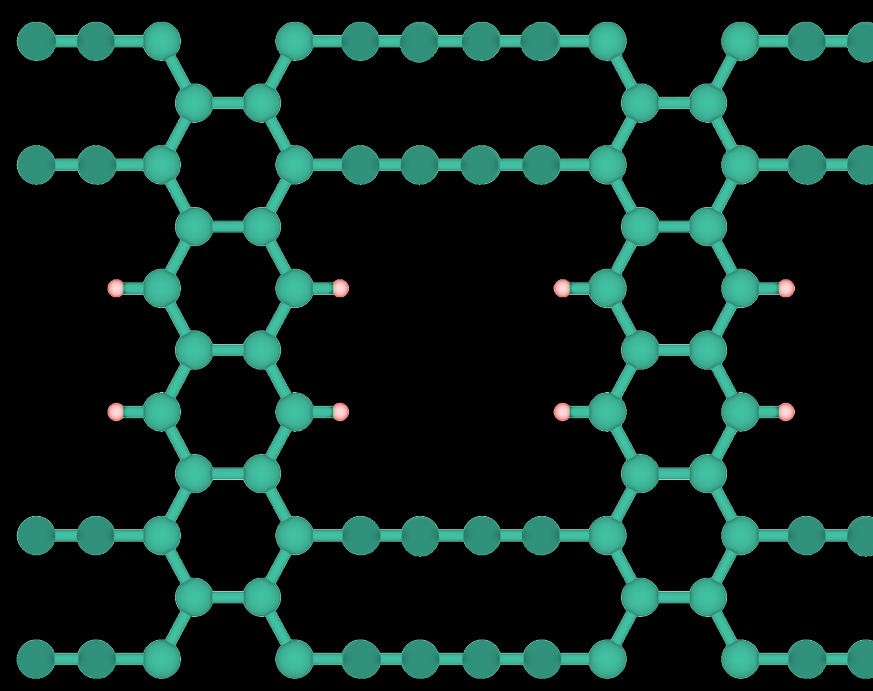


## GRAZYNE MODELS



[1],[2]{2} ⇌ A

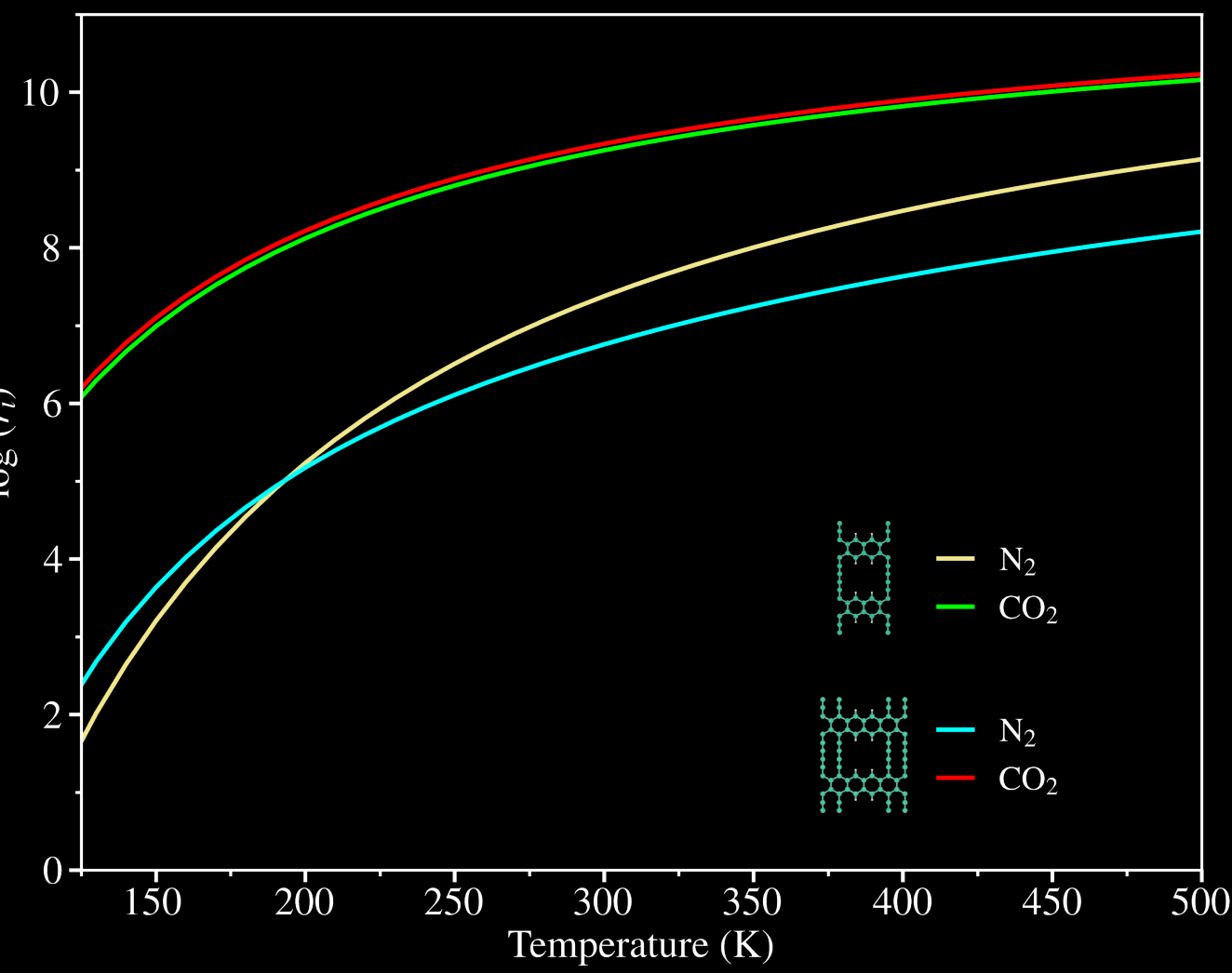
Composed of one-ring wide graphene stripes interconnected by acetylenic bonds, spanning a distance equivalent to two triple bonds, with two acetylenic vacancies.



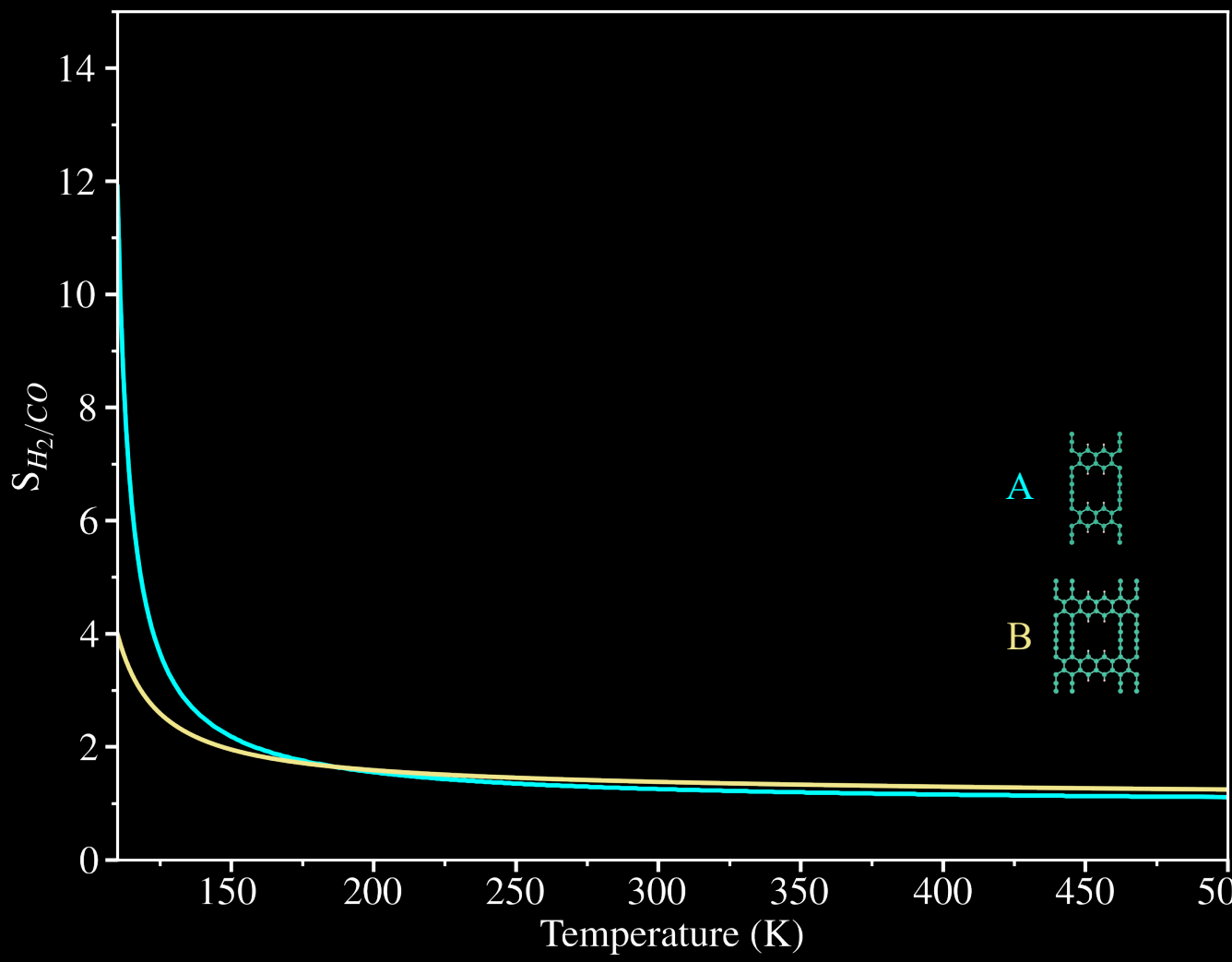
[1],[2]{(0,0),2} ⇌ B

The difference between the structures lies in the A-grazyne containing a solitary acetylenic bond between consecutive pores, while the B-grazyne exhibits two adjacent acetylenic bonds followed by two vacancies.

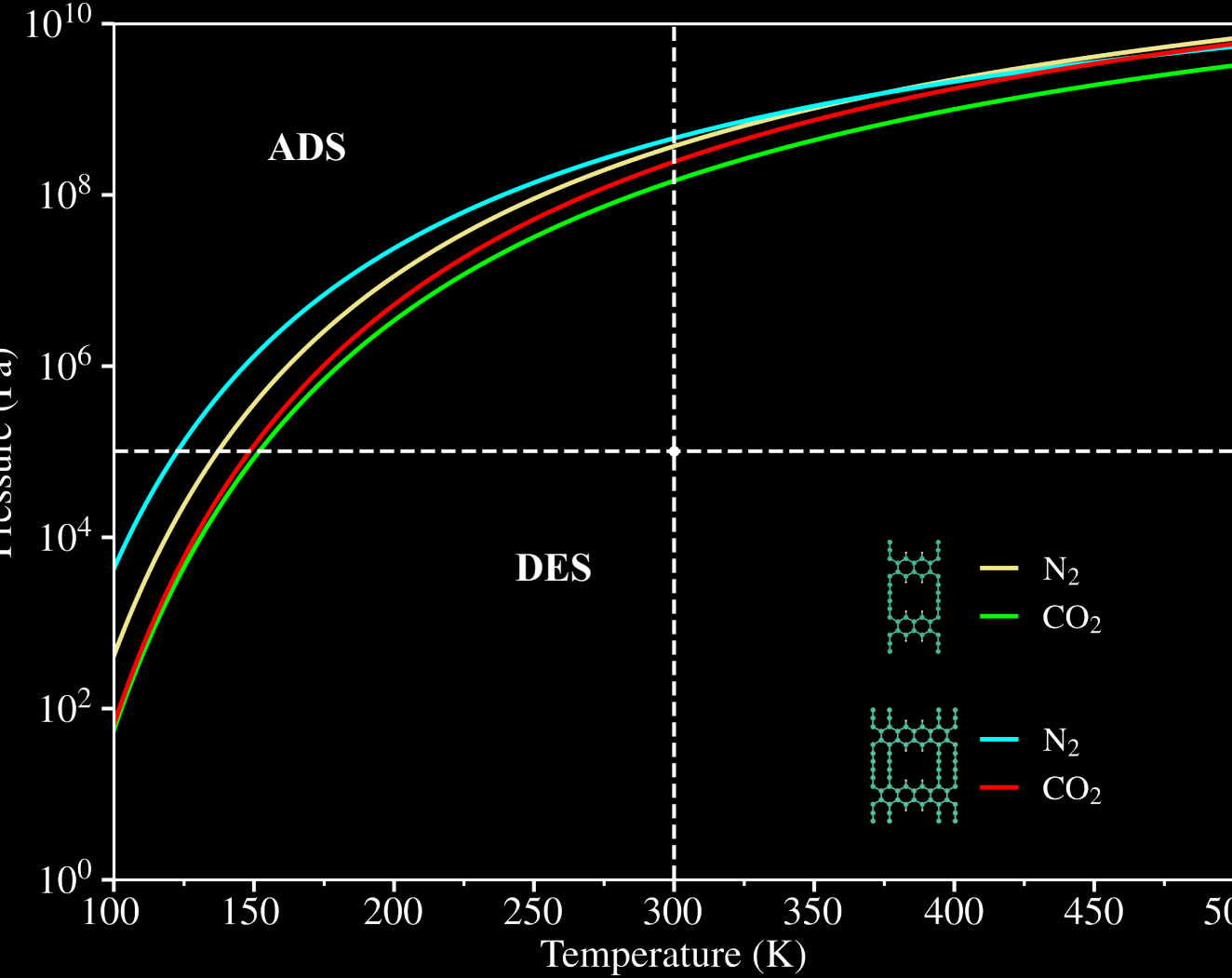
## DFT RESULTS



CO<sub>2</sub> consistently shows higher membrane diffusion rates than N<sub>2</sub>, regardless of the crazyne model. The figure highlights a key temperature range between 100 and 500 K, where diffusion rates,  $r_i$ , exceed 1 s<sup>-1</sup>, suggesting the potential use of grazynes as molecular separation membranes.



The analysis of the selectivity confirms that CO<sub>2</sub> has a higher selectivity than N<sub>2</sub>, but this decreases as temperature goes up, following the rate constants, with a clear trend toward unity at higher temperatures. Therefore, lower temperatures are better for separating CO<sub>2</sub> from N<sub>2</sub>, though this comes with a lower permeation rate.



The KPD highlights two distinct regions: one where adsorption is favored, and another where desorption is preferred. CO<sub>2</sub> shows a wider adsorption range. At p=1 atm and T=300 K, neither CO<sub>2</sub> nor N<sub>2</sub> is inclined toward adsorption, making grazynes effective for separation membranes.

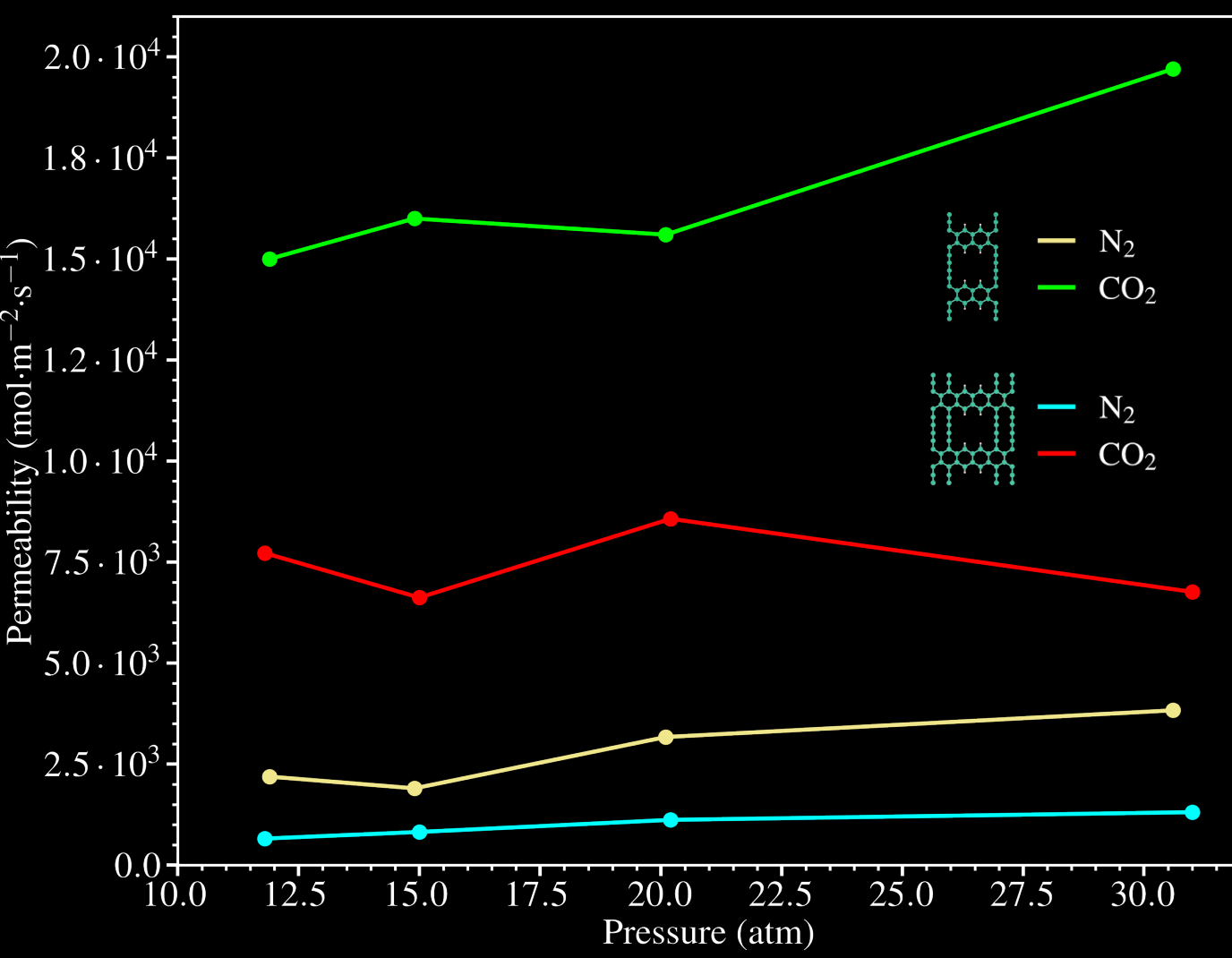
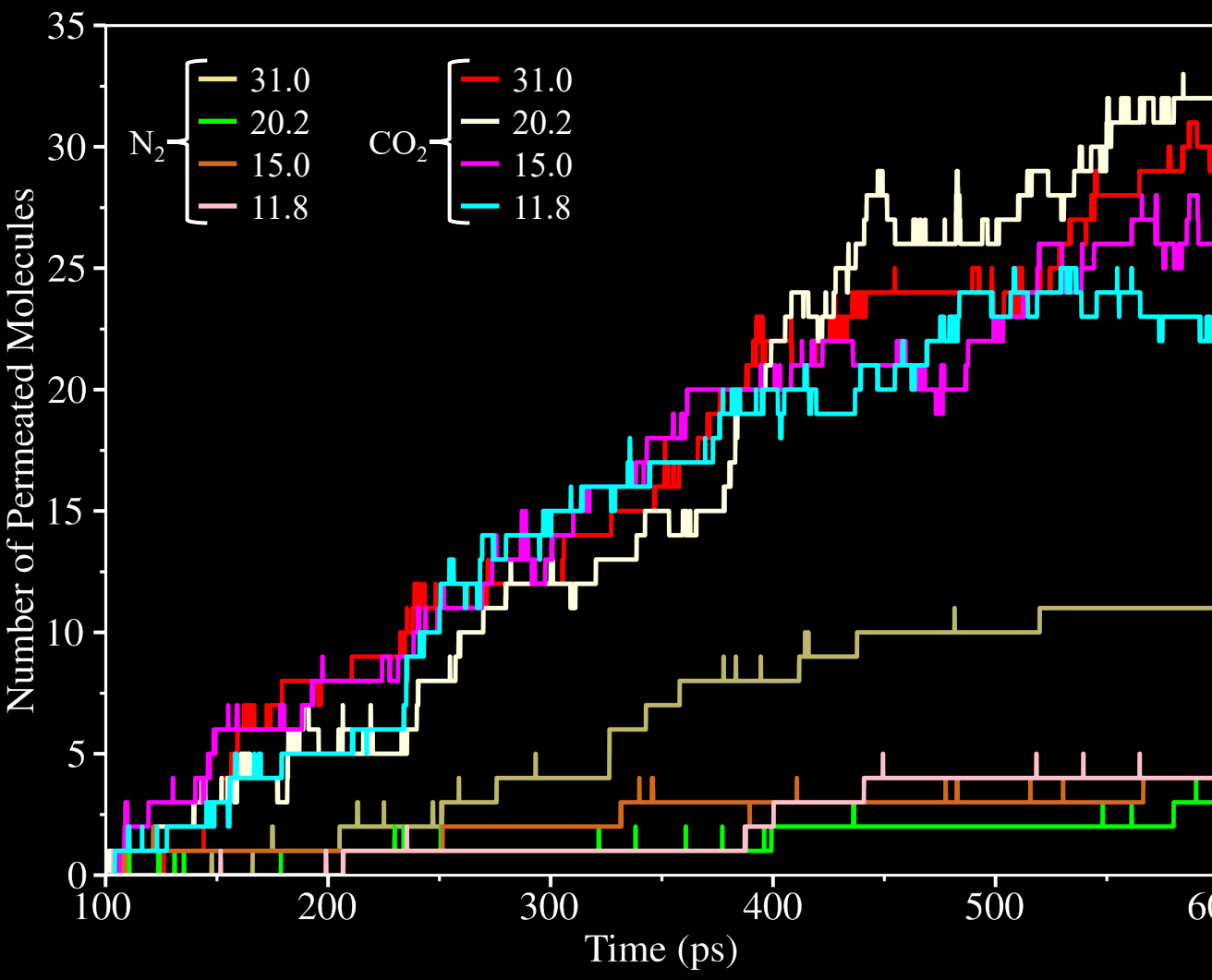
## CONCLUSIONS

DFT and MD support the use of crazyne membranes for selective CO<sub>2</sub> separation from N<sub>2</sub>. Grazynes are capable of physisorbing CO<sub>2</sub> and N<sub>2</sub>, thus avoiding material poisoning by molecular decoration, while the diffusion of CO<sub>2</sub> through the pores is found to be rapid, yet easier than that of N<sub>2</sub>, in the rate order of the s<sup>-1</sup> in the 100-500 K temperature range. In addition, according to MD with equimolecular mixtures of CO<sub>2</sub>:N<sub>2</sub>, ca. 60-70% of CO<sub>2</sub> permeates the A-grazyne, with the remaining 30-40% being N<sub>2</sub>. Conversely, B-grazyne achieves up to 90% CO<sub>2</sub> permeation.

## MOLECULAR DYNAMICS RESULTS

| Grazyne          | A    | B    |
|------------------|------|------|
| Pressure (atm)   | 30.6 | 20.1 |
| 500 ps           | 14.9 | 11.9 |
| 1500 ps          | 31.0 | 20.2 |
| #N <sub>2</sub>  | 18   | 14   |
| #CO <sub>2</sub> | 39   | 41   |
| #N <sub>2</sub>  | 31   | 31   |
| #CO <sub>2</sub> | 41   | 38   |

The number of CO<sub>2</sub> (#CO<sub>2</sub>) and N<sub>2</sub> (#N<sub>2</sub>) molecules that passed through shows that CO<sub>2</sub> diffuses more easily through both crazyne membranes. CO<sub>2</sub> makes up about 65-75% of the total molecules permeating through A-grazyne and 70-90% for B-grazyne. These results suggest a potential CO<sub>2</sub> enrichment of ca. 90%. This trend is also seen in the time evolution of the number of molecules that passed through B-grazyne, while a similar trend is observed for A-grazyne (not shown). Generally, increasing the pressure leads to more permeation, though some trends may fluctuate. It is important to remember that the likelihood of a molecule passing through a pore depends on its arrival orientation. No parallel permeation has been observed, thus molecules approaching perpendicularly have a higher chance of crossing.



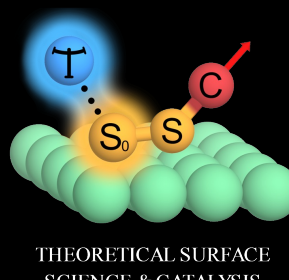
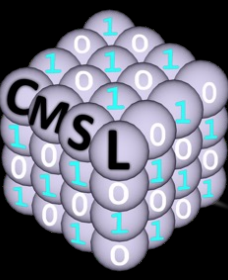
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S. Kamalinahad, F. Viñes, P. Gamallo, *J. Phys. Chem. C* **2019**, 123, 27140–27149.  
F. Viñes, A. Calzada, P. Gamallo, *J. CO<sub>2</sub> Util.* **2023**, 71, 102459.

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